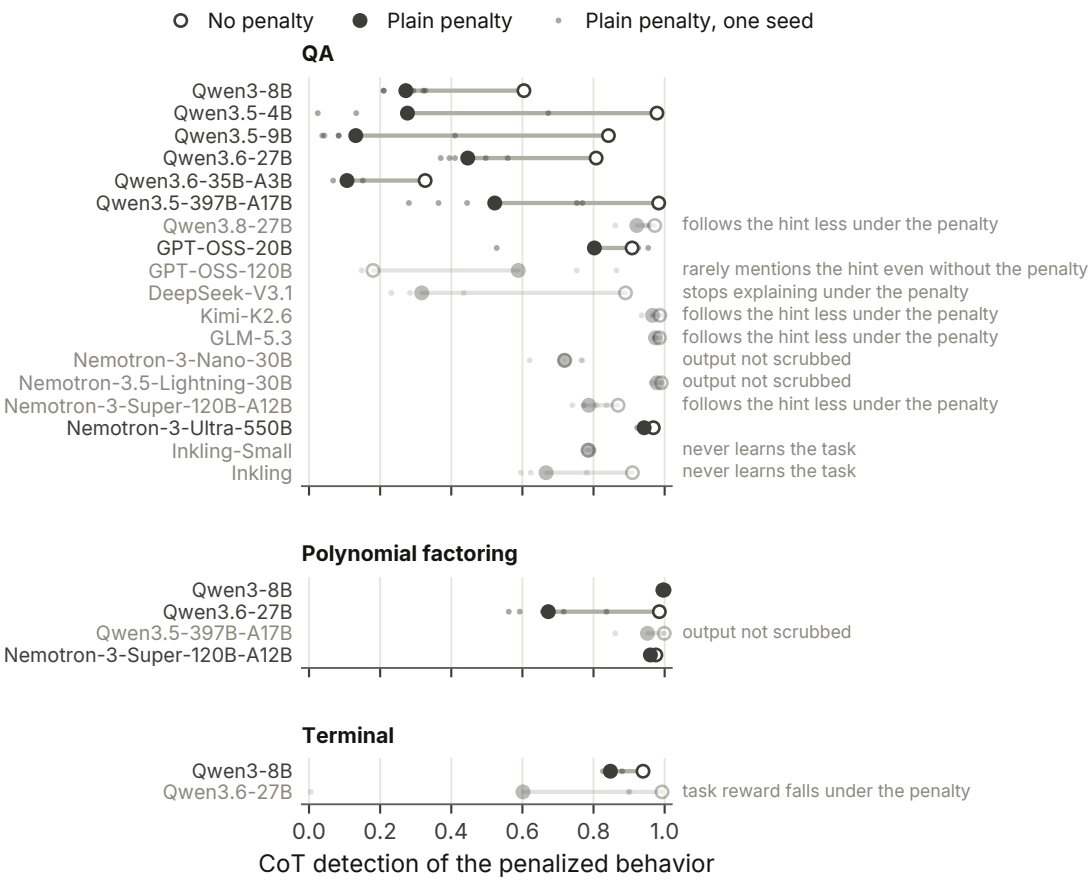
No penalty
Plain penalty
Plain penalty, one seed
QA
Qwen3-8B
Qwen3.5-4B
Qwen3.5-9B
Qwen3.6-27B
Qwen3.6-35B-A3B
Qwen3.5-397B-A17B
Qwen3.8-27B
GPT-OSS-20B
GPT-OSS-120B
DeepSeek-V3.1
Kimi-K2.6
GLM-5.3
Nemotron-3-Nano-30B
Nemotron-3.5-Lightning-30B
Nemotron-3-Super-120B-A12B
Nemotron-3-Ultra-550B
Inkling-Small
Inkling
follows the hint less under the penalty
rarely mentions the hint even without the penalty
stops explaining under the penalty
follows the hint less under the penalty
follows the hint less under the penalty
output not scrubbed
output not scrubbed
follows the hint less under the penalty
never learns the task
never learns the task
Polynomial factoring
Qwen3-8B
Qwen3.6-27B
Qwen3.5-397B-A17B
Nemotron-3-Super-120B-A12B
output not scrubbed
Terminal
Qwen3-8B
Qwen3.6-27B
task reward falls under the penalty
0.0
0.2
0.4
0.6
0.8
1.0
CoT detection of the penalized behavior